Robust Code-Switched Lecture Transcription and Cross-Lingual Voice Cloning Pipeline for Low-Resource Maithili

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GitHub Repository: Link to Code

***Abstract*—This research addresses the critical gap in educational accessibility for the Maithili-speaking population by developing an end-to-end pipeline capable of transcribing "Hinglish" (Hindi-English code-switched) academic lectures and synthesizing them into Maithili using the original speaker's voice. Our approach tackles several low-resource challenges, including acoustic noise in lecture halls, the lack of parallel corpora for Maithili, and the complexity of preserving prosody across different languages. We implement a multi-stage system featuring multi-band spectral subtraction, a dual-head CNN for language identification (LID), and a VITS-based synthesis engine with DTW-driven prosody warping. The codebase and datasets developed for this study are made available to the research community to foster further development in Indic low-resource speech tasks. We further implement LFCC-based anti-spoofing countermeasures and perform adversarial robustness analysis using FGSM. Comprehensive evaluations demonstrate a Word Error Rate (WER) of under 15% for English segments, a Mel-Cepstral Distortion (MCD) below 8.0 for synthesis, and an Equal Error Rate (EER) of less than 10% for the anti-spoofing classifier.**

***Index Terms*—Code-Switching, Maithili, Zero-Shot Voice Cloning, Language Identification, N-gram Logit Biasing, Dynamic Time Warping, Anti-Spoofing, Adversarial Robustness.**

I. Introduction

The linguistic landscape of the Indian subcontinent is characterized by a unique phenomenon known as code-switching. In higher education, particularly in STEM fields, instructors rarely stick to a single language. Instead, they use a fluid mixture of English (for technical terminology) and Hindi (for conceptual scaffolding). While this "Hinglish" discourse is effective for bilingual students, it creates a massive barrier for students from regional linguistic backgrounds, such as Maithili speakers.

Maithili is an Indo-Aryan language spoken by over 34 million people primarily in the Indian state of Bihar and the eastern Terai region of Nepal. Despite its large speaker base, it remains "low-resource" in the digital world. There is a scarcity of high-quality labeled speech data, making it difficult to train modern neural Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) systems from scratch.

This paper details a "cross-lingual" strategy to bridge this gap: we leverage the high-resource data of Hindi and English to build a pipeline to Maithili. Furthermore, by using zero-shot voice cloning, we ensure the translated lecture sounds like the original professor. Preserving the speaker's original voice, intonation, and prosodic cues has been shown to increase student engagement, maintain the pedagogical emphasis, and build trust in the digital content.

However, building such a pipeline introduces several significant technical challenges:

1) **Acoustic Disturbance:** Real-world classroom recordings are plagued by reverberation, background chatter, and non-stationary noises (e.g., AC hums, projector fans).
2) **Code-Switched Decoding:** Standard ASR systems are heavily biased toward monolingual decoding. They often misrecognize English technical terms embedded in Hindi sentences due to altered phonetic pronunciations.
3) **Translation Bottlenecks:** There is virtually no standard technical parallel corpus mapping complex English/Hinglish engineering terms to Maithili.
4) **Prosodic Transfer:** Transferring speech characteristics across languages with different syntactic structures (e.g., Subject-Verb-Object vs. Subject-Object-Verb) typically breaks temporal alignments, leading to unnatural, robotic speech.
5) **Security Vulnerabilities:** The ease of generating synthetic voices introduces security risks, necessitating robust anti-spoofing countermeasures and analysis against adversarial attacks.

To address these challenges, we propose a comprehensive, multi-stage pipeline. The rest of this paper is organized as follows: Section II reviews related work. Section III details the methodology for Part I (Transcription and LID). Section IV discusses Part II (Phonetic Mapping and Translation). Section V covers Part III (Voice Cloning and Synthesis). Section VI explores Part IV (Security and Robustness). Section VII details the System Architecture and Algorithms. Section VIII presents the experimental setup, evaluation metrics, and comprehensive results, including ablation studies. Section IX discusses limitations and broader impacts, Section X provides

code availability, and Section XI concludes the paper.

II. Related Work

A. Code-Switched Speech Recognition

Code-switched ASR has gained significant attention in recent years. Early approaches relied on combining multiple monolingual acoustic and language models. However, modern end-to-end models like wav2vec 2.0 [1] and Whisper [2] have shown remarkable zero-shot capabilities across languages. Despite these advances, Whisper struggles with domain-specific terminology in heavily code-switched environments without explicit contextual biasing. The phonetic boundaries in Hinglish are highly blurred, causing catastrophic forgetting of minority language grammar during autoregressive generation.

B. Cross-Lingual Voice Cloning

Voice cloning aims to synthesize speech in a target speaker's voice using only a few seconds of reference audio. Systems like YourTTS [3] and VITS [4] have demonstrated impressive zero-shot multi-speaker and multilingual capabilities. The primary challenge in cross-lingual cloning is preserving the source speaker's prosody (pitch and rhythm) while speaking a language they may not actually know, which requires explicit prosody modeling and alignment techniques like Dynamic Time Warping (DTW). Previous works have largely focused on high-resource European languages, leaving Indo-Aryan tonal and stress patterns underexplored.

C. Low-Resource Language Processing

Maithili computational linguistics is still in its infancy. Most existing work focuses on basic rule-based Machine Translation or morphological analysis. Neural machine translation (NMT) models like NLLB (No Language Left Behind) [5] support Maithili but often fail on highly technical domain-specific terminology, necessitating the creation of custom parallel corpora. The lack of standard Grapheme-to-Phoneme (G2P) dictionaries further complicates TTS synthesis for Maithili, requiring hybrid rule-based and statistical fallback methods.

D. Audio Anti-Spoofing and Adversarial Robustness

The ASVspoof challenges [6] have driven the development of countermeasures against synthetic speech. Linear Frequency Cepstral Coefficients (LFCC) combined with Gaussian Mixture Models (GMM) remain a strong baseline for detecting vocoder artifacts. Simultaneously, adversarial attacks like the Fast Gradient Sign Method (FGSM) [7] have been adapted to the audio domain, revealing the fragility of neural acoustic models. Protecting educational speech pipelines from targeted prompt injection and voice spoofing is becoming a critical subfield of audio forensics.

III. Phase I: Robust Code-Switched Transcription (STT)

The first phase of our pipeline transforms the raw, noisy classroom audio into a highly accurate, language-tagged text transcript.

A. Task 1.3: Acoustic Preprocessing and Denoising

Lecture recordings are notoriously difficult to process. They often contain non-stationary environmental noise. Basic spectral subtraction often introduces "musical noise" artifacts due to uniform over-subtraction. To clean the signal effectively, we implemented a multi-band frequency-dependent spectral subtraction algorithm.

The audio signal is decomposed into multiple frequency bands using a filter bank. For each band, we calculate a specific noise over-subtraction factor, $\alpha(\omega)$.

$$|\hat{S}(\omega)|^2 = \max\left(|Y(\omega)|^2 - \alpha(\omega)|\hat{N}(\omega)|^2, \beta|\hat{N}(\omega)|^2\right) \quad (1)$$

Where:

- $Y(\omega)$ is the magnitude spectrum of the noisy signal.
- $\hat{N}(\omega)$ is the estimated noise spectrum, continuously updated during periods of silence (detected via energy thresholds).
- β is the spectral floor parameter (set to 0.002) to prevent the spectrum from dropping to zero, which causes unnatural artifacts.

In this implementation, the value of α is tuned based on the characteristics of a typical classroom:

- **Low Frequencies (0-300Hz):** $\alpha = 3.0$. This heavily suppresses AC hum and electronic droning.
- **Mid Frequencies (300Hz-4kHz):** $\alpha = 1.5$. This gentle subtraction preserves the vital formants of the human voice, which are crucial for intelligibility.
- **High Frequencies ($>$ 4kHz):** $\alpha = 1.2$. This reduces subtraction to prevent the loss of high-frequency fricatives (e.g., /s/, /f/, /sh/).

Following spectral subtraction, we apply a dereverberation filter using an exponential moving average to estimate and subtract late reflections, improving the signal-to-reverberant ratio.

B. Task 1.1: Multi-Head Language Identification (LID)

Before generating text, the system must decide which language "mode" it is in at the frame level. Code-switching boundaries in Hinglish can occur mid-sentence or even mid-word. Standard LID systems trained on monolingual utterances fail here.

We developed a dual-head Convolutional Neural Network (CNN) that processes 500ms audio frames (with 250ms overlap) converted into 80-dimensional log-mel spectrograms.

Architecture Design: The shared trunk consists of three 2D Convolutional layers with Batch Normalization and ReLU activations, followed by Max Pooling. This trunk learns foundational acoustic representations. The network then splits into two specialized heads:

1) **Spectral Head:** Focuses on the distribution of formants and phonetic features along the frequency axis.
2) **Temporal Head:** Focuses on rhythmic and syllabic timing along the time axis.

Mathematical Formulation of Ensemble: Given the shared feature map F, the logits from the spectral head Z_S and temporal head Z_T are computed. The final frame-level language probability distribution $P(L)$ is the averaged softmax output:

$$P(L|X) = \frac{1}{2}\left(\sigma(W_S F + b_S) + \sigma(W_T F + b_T)\right) \quad (2)$$

By combining these two heads, the model cross-validates prosodic cues against phonetic cues. This drastically reduces false language switches at borrowed terms (e.g., an English technical term pronounced with a Hindi accent), achieving a target F1-score of > 0.85.

C. Task 1.2: Constrained Decoding via Logit Biasing

The core transcription engine utilizes OpenAI's Whisper-v3. While Whisper is highly robust, it suffers from a critical flaw in academic settings: it frequently misspells domain-specific technical terminology, especially when heavily accented.

Instead of hard-constraining the beam search (which restricts vocabulary entirely and breaks on conversational Hindi words), we implemented a soft N-gram logit biasing layer.

N-gram Language Model Construction: We compiled a domain-specific corpus from a standard Speech Processing syllabus, extracting over 200 technical terms (e.g., "stochastic", "cepstrum", "formant", "Viterbi"). We trained a trigram language model with Kneser-Ney smoothing on this text.

Decoding Modification: During Whisper's autoregressive decoding step, we intercept the raw output logits z_w for every token w in the vocabulary. Let the current decoded context be $c = (w_{i-n+1}, \ldots, w_{i-1})$. We compute a modified logit z'_w:

$$z'_w = z_w + \lambda \cdot \log P_{\text{ngram}}(w|c) + \lambda_t \cdot \mathbb{1}[w \in \mathcal{T}] \quad (3)$$

Where:

- $\lambda = 2.5$ is the baseline bias weight.
- P_{ngram} is the probability from our custom trigram model.
- $\mathcal{T}$ is the predefined set of critical technical terms.
- $\lambda_t = 3.75$ (which is $1.5 \times \lambda$) provides an aggressive additive boost to ensure technical terms are prioritized even if acoustic confidence is low.

This approach allows the decoder to freely generate conversational Hindi while strongly steering it toward correct spellings of technical jargon when the acoustic context suggests a technical term is imminent.

IV. Phase II: Phonetic Mapping and Translation

Once an accurate Hinglish transcript is obtained, it must be translated to the target low-resource language (Maithili). This involves bridging phonetic gaps and semantic translation.

A. Task 2.1: IPA Unified Representation

Standard Grapheme-to-Phoneme (G2P) mappers assume monolingual text. Feeding Hinglish text into an English G2P model results in catastrophic mispronunciations of Hindi words. Conversely, Hindi G2P models fail on English acronyms and spellings.

To solve this, we implemented a custom, LID-routed G2P mapping algorithm to convert the transcript into a unified International Phonetic Alphabet (IPA) string.

Routing Logic: For each word in the transcript, we align its temporal boundaries with the frame-level LID predictions.

- If the word falls predominantly in a "Hindi" frame, it is routed to a custom Devanagari/Roman-Hindi to IPA rule engine.
- If the word falls in an "English" frame, it is routed to Epitran (an English G2P library) with rule-based fallback.

Hinglish Bridging Dictionary: We created a specialized lookup table for high-frequency Hinglish discourse markers that confuse standard models. For example, the word "matlab" (meaning "that is" or "meaning" in Hindi) is mapped to /mətləb/ rather than English phonetic rules attempting to pronounce "mat-lab". Similarly, discourse words like "accha", "toh", and "lekin" are explicitly mapped.

B. Task 2.2: Semantic Translation and Parallel Corpus

Translating the transcript to Maithili presents a "zero-resource" problem for specific technical jargon. State-of-the-art MT models like NLLB-200 support Maithili but output nonsensical translations or hallucinate when encountering words like "spectrogram" or "convolution".

To overcome this, we constructed a custom 500+ word parallel corpus mapping English/Hinglish speech processing terminology directly to Maithili in the Devanagari script.

Translation Strategy:

1) **N-gram Corpus Lookup:** We greedily search the transcript for unigrams, bigrams, and trigrams present in our technical parallel corpus and directly substitute the Maithili equivalents.
2) **Stemming Fallback:** If a word is not found, we apply basic English stemming (removing -ing, -tion, -ed) and re-query the corpus.
3) **NLLB Fallback:** For the remaining conversational "glue" words and phrases not covered by our technical dictionary, we pass the segments to the NLLB-200-distilled-600M neural translation model.

This hybrid approach ensures semantic accuracy for critical engineering concepts while maintaining natural grammatical flow for the surrounding lecture discourse.

V. Phase III: Zero-Shot Cross-Lingual Voice Cloning

This phase synthesizes the translated Maithili text into audio, preserving the original professor's unique voice and teaching style.

A. Task 3.1: Voice Embedding Extraction

To clone the voice, the system requires a mathematical representation of the target speaker's vocal characteristics, independent of the words being spoken.

We utilize a pre-trained ECAPA-TDNN (Emphasized Channel Attention, propagation and aggregation in TDNN) model to extract a 192-dimensional x-vector from a 60-second reference audio of the student/professor.

The ECAPA-TDNN architecture is highly robust for speaker verification and captures fine-grained vocal tract characteristics. If neural extraction fails, our system employs a statistical fallback mechanism, calculating the mean, standard deviation, and delta features across 80 mel-frequency bands to create a pseudo-d-vector.

B. Task 3.2: Prosody Warping via Dynamic Time Warping (DTW)

A major flaw in naive voice cloning is that while the synthesized audio sounds like the target speaker, it lacks their original emotion, rhythm, and emphasis. This is particularly problematic in education, where pitch variations indicate important concepts. We must transfer the prosody (pitch and energy) from the original Hinglish audio to the synthesized Maithili audio.

However, translated sentences have different lengths and phonetic timings. We use Dynamic Time Warping (DTW) to align the two temporal sequences.

Feature Extraction: We extract the fundamental frequency (F_0) contour using the pYIN algorithm and calculate the frame-level RMS energy.

FastDTW Alignment: Standard DTW has an $O(N^2)$ complexity, which is computationally prohibitive for a 10-minute lecture (resulting in a matrix of size $60,000 \times 60,000$ frames). We utilize FastDTW, an approximation algorithm that operates in $O(N)$ time.

To prevent pathological warping (where one frame of the original audio is stretched over an entire synthesized sentence), we impose a Sakoe-Chiba band constraint ($r = 50$ frames, equivalent to 500ms):

$$D(i,j) = d(x_i, y_j) + \min \begin{cases} D(i-1,j) \\ D(i,j-1) \\ D(i-1,j-1) \end{cases} \tag{4}$$

Subject to: $|i - j| \leq r$

F0 Normalization and Warping: The aligned F_0 contour of the professor must be adapted to the physical vocal range of the target speaker (student's reference voice) to prevent synthesis artifacts. We apply z-score normalization:

$$\hat{f}_0^{\text{synth}}(t) = \left(\frac{f_0^{\text{prof}}(t) - \mu_{\text{prof}}}{\sigma_{\text{prof}}} \right) \cdot \sigma_{\text{ref}} + \mu_{\text{ref}} \tag{5}$$

Where μ and σ represent the mean and standard deviation of the voiced F_0 segments. The warped energy and pitch contours are then used to modulate the final synthesized waveform.

C. Task 3.3: Synthesis Generation

The final synthesis is performed using a VITS-based architecture adapted for multilingual zero-shot TTS (e.g., YourTTS). VITS combines Variational Autoencoders (VAEs), normalizing flows, and adversarial training to generate highly natural speech.

The model takes the translated Maithili text, the extracted x-vector speaker embedding, and the warped prosodic contours to generate the final waveform. The output is strictly formatted to a sampling rate of 22.05 kHz or higher to ensure high-fidelity audio suitable for academic listening.

VI. Phase IV: Adversarial Robustness and Spoofing Detection

As voice cloning technologies become democratized, the threat of malicious audio spoofing (deepfakes) increases. Furthermore, acoustic models (like our LID) are vulnerable to adversarial noise. We address both security aspects.

A. Task 4.1: Anti-Spoofing Classifier (Countermeasure)

We implemented a Countermeasure (CM) system to classify audio as "Bona Fide" (real human speech) or "Spoof" (synthesized by our Phase III pipeline).

Feature Extraction: Neural vocoders (like HiFi-GAN used in VITS) often leave microscopic artifacts in the high-frequency spectrum. Standard Mel-Frequency Cepstral Coefficients (MFCCs) use a log-spaced filterbank that compresses high frequencies, effectively hiding these artifacts. Therefore, we utilize Linear Frequency Cepstral Coefficients (LFCCs), which apply a linearly spaced filterbank, preserving high-frequency vocoder anomalies. We extract 20 LFCCs, along with their Δ and $\Delta\Delta$ coefficients, resulting in a 60-dimensional feature vector per frame.

GMM Backend: Given the limited amount of bona fide reference data (60 seconds) compared to the spoofed output (10 minutes), deep neural classifiers would severely overfit. We employ Gaussian Mixture Models (GMMs) with 16 diagonal-covariance components. We train two separate GMMs: GMM_{bona} and $\text{GMM}_{\text{spoof}}$.

The final decision score for an unknown audio sequence X is the log-likelihood ratio:

$$\Lambda(X) = \frac{1}{T} \sum_{t=1}^{T} \left(\log P(x_t | \text{GMM}_{\text{bona}}) - \log P(x_t | \text{GMM}_{\text{spoof}}) \right) \tag{6}$$

A positive score indicates bona fide speech, while a negative score indicates a spoof.

B. Task 4.2: Adversarial Noise Injection (FGSM)

To test the robustness of our Phase I LID system, we perform an adversarial attack using the Fast Gradient Sign Method (FGSM). The goal is to find an imperceptible noise perturbation that forces the LID to misclassify a Hindi frame as English.

We compute gradients in the log-mel spectrogram domain with respect to the LID loss function $\mathcal{L}$:

$$X_{\text{adv}} = X + \epsilon \cdot \text{sign}\left(\nabla_X \mathcal{L}(\theta, X, y_{\text{target}}) \right) \tag{7}$$

Where X is the input mel-spectrogram, y_{target} is the incorrect language label (English), and ϵ controls the perturbation magnitude.

To ensure the noise remains inaudible to humans, we enforce a strict Signal-to-Noise Ratio (SNR) constraint of > 40 dB. We implement a binary search algorithm to find the minimum ϵ that successfully flips the prediction while strictly satisfying the SNR constraint. This perturbation is then projected back from the mel-spectrogram to the raw audio waveform using the pseudo-inverse of the mel basis matrix and an inverse STFT.

VII. System Architecture and Algorithms

In this section, we present the formal algorithmic pseudocode that drives the core components of our pipeline. This ensures complete reproducibility of our cross-lingual bridging techniques.

Algorithm 1 Logit-Biased Constrained Decoding

Require: Audio feature map X, Technical Corpus $\mathcal{T}$, Bias weight λ, Trigram model P_{ngram}

```
Initialize decoder sequence S ← [BOS]
while token ≠ EOS do
    Z ← WhisperDecoder(S, X) {Raw logits}
    C ← ExtractRecentTrigram(S)
    for each token w in Vocabulary do
        Z'_w ← Z_w + λ log P_ngram(w|C)
        if w ∈ T then
            Z'_w ← Z'_w + 1.5λ {Aggressive technical boost}
        end if
    end for
    w_next ← argmax(Z')
    Append w_next to S
end while
return S
```

Algorithm 2 Sakoe-Chiba Constrained FastDTW Prosody Warping

Require: Professor F_0 contour $P \in \mathbb{R}^N$, Synthesized F_0 contour $S \in \mathbb{R}^M$, Bandwidth constraint r

```
Initialize Cost Matrix D_{N×M} ← ∞
D_{0,0} ← 0
for i = 1 to N do
    for j = max(1, i − r) to min(M, i + r) do
        cost ← |P_i − S_j|
        D_{i,j} ← cost + min(D_{i−1,j}, D_{i,j−1}, D_{i−1,j−1})
    end for
end for
Backtrack from D_{N,M} to D_{0,0} to find optimal warp path W
return W
```

The integration of these algorithms ensures that both the linguistic meaning (via Algorithm 1) and the emotional/pedagogical emphasis (via Algorithm 2) are preserved during the cross-lingual transition.

VIII. Experimental Setup, Results, and Evaluation

A. Datasets and Implementation Details

The pipeline was evaluated on a provided 10-minute code-switched classroom recording. The student reference voice was generated/recorded for 60 seconds. The pipeline is implemented in Python using PyTorch, Librosa, and Transformers. Training and inference were executed on standard consumer-grade hardware equipped with an NVIDIA RTX 3090 GPU to accelerate the Whisper-v3 and VITS inference passes.

Hyperparameter tuning for the Multi-band Spectral Subtraction was conducted empirically over 50 varied 5-second noise segments extracted from silent periods in the lecture audio.

B. LID Performance and Confusion Matrix

The dual-head LID system was evaluated on manually annotated frames. Table I presents the confusion matrix.

TABLE I: Code-Switching LID Confusion Matrix (Frame-Level)

	Predicted English	**Predicted Hindi**
True English	580 (TP)	34 (FN)
True Hindi	45 (FP)	1739 (TN)

The system achieved a precision of 0.928 and a recall of 0.944 for English, yielding an F1-score of > 0.85. The majority of False Positives occurred during extremely short Hindi conjunctions bounded by English technical terms. The timestamp precision for language switches was well within the required 200ms tolerance.

C. Transcription Word Error Rate (WER)

The implementation of the N-gram logit biasing significantly improved the recognition of technical jargon.

- **English Segments WER:** 12.4% (Passes $< 15\%$ criteria).
- **Hindi Segments WER:** 21.8% (Passes $< 25\%$ criteria).

Without logit biasing, the English WER was 18.2%, highlighting the effectiveness of the targeted probability boosting for syllabus terms.

D. Translation Quality Analysis

Evaluating Maithili translation quantitatively via BLEU scores is unreliable due to the absence of a standardized gold-reference dataset for speech processing terminology in Maithili. Consequently, we conducted a qualitative human-in-the-loop review for 100 random technical sentences. Our semantic translation approach (using the 500-word parallel corpus prior to NLLB fallback) successfully retained the engineering meaning in 92% of the test cases. When relying solely on NLLB-200 without the dictionary, semantic retention dropped to 41%, with the neural model frequently transliterating English terms arbitrarily or generating phonetically similar but semantically meaningless Maithili words.

E. Ablation Study: Prosody Warping

To quantify the impact of DTW-based prosody warping, we conducted an ablation study measuring Mel-Cepstral Distortion (MCD) and human Mean Opinion Score (MOS) for naturalness. MCD measures the objective acoustic distance between the reference voice and the synthesized voice.

TABLE II: Ablation Study on Prosody Warping

Method	MCD ↓	Naturalness (MOS) ↑
Baseline (Flat Synthesis)	9.4	2.1
Pitch (F_0) Warping Only	8.1	2.9
Energy Warping Only	8.7	2.5
Full DTW (F0 + Energy)	**7.2**	**3.6**

As shown in Table II, implementing the full DTW prosody warping reduced the MCD to 7.2 (passing the < 8.0 strict criteria) and significantly improved the subjective naturalness, eliminating the robotic cadence of the baseline synthesizer.

F. Security Evaluation

Anti-Spoofing: The LFCC-GMM classifier was evaluated on a test set comprising the bona fide reference audio and various segments of the synthesized output. The system achieved an **Equal Error Rate (EER) of 6.8%**, successfully passing the $< 10\%$ requirement. The linear frequency scaling proved highly effective at isolating the high-frequency artifacts generated by the VITS vocoder.

Adversarial Robustness: The binary search for the FGSM attack revealed that the LID system is relatively robust. The minimum perturbation ϵ required to flip a Hindi prediction to English while maintaining an SNR > 40 dB was $\epsilon = 0.018$. Perturbations smaller than this failed to alter the dual-head consensus, while larger perturbations became perceptually audible (violating the SNR constraint).

IX. Discussion and Limitations

The proposed pipeline successfully navigates the complex transition from a noisy, code-switched Hinglish lecture to a clean, prosodically matched Maithili synthesis. The logit biasing approach proved to be a computationally cheap yet highly effective method for domain adaptation without retraining massive ASR weights.

A. Sociolinguistic Impact

By providing educational material in a student's native tongue, we significantly reduce the cognitive load required to parse foreign languages, allowing students to focus purely on the technical engineering concepts. This holds massive potential for democratizing technical education across rural India, where Maithili is widely spoken but rarely used as a medium of higher instruction.

B. Technical Limitations

The pipeline relies heavily on the quality of the manually curated parallel corpus for technical terms. Scaling this approach to cover entire university degree programs would require significant lexicographical effort to standardize Maithili engineering terminology. Furthermore, while the DTW prosody transfer works well for short to medium sentences, highly complex syntactic reorderings during translation can cause the warped pitch contour to align unnaturally with the target syllables. Subject-Object-Verb (Hindi/Maithili) to Subject-Verb-Object (English) translation inherently alters the timing of emphasis.

X. Code Availability

To support reproducibility and encourage further development in low-resource Indic speech processing, the complete source code for the denoising modules, custom Whisper Logit-Biased decoders, DTW prosody warpers, and the LFCC-GMM anti-spoofing countermeasures has been made publicly available.

The repository includes the 500+ word Maithili parallel corpus and the Hinglish G2P bridging dictionaries developed exclusively for this study.

GitHub Repository: https://github.com/b22ai049/Speech-Assignment-2

XI. Conclusion

This research demonstrates the viability of utilizing state-of-the-art zero-shot AI models combined with targeted classical DSP techniques (spectral subtraction, DTW, LFCC) to solve real-world educational accessibility issues. By delivering a complete end-to-end pipeline, we proved that it is possible to accurately transcribe Hinglish lectures and synthesize them into low-resource Maithili while preserving the original instructor's acoustic identity. The adherence to strict evaluation criteria—including WER, MCD, and anti-spoofing EER—ensures the system's robustness and security. The provided codebase and parallel corpus aim to accelerate further research in inclusive educational technologies for the Indian subcontinent.

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